PRODUCT OVERVIEW

BrickTrace

See how your data really flows — and act on it.

BrickTrace turns Unity Catalog's raw lineage into an interactive map of your entire data estate: every table, every column, the exact logic that transformed it, who can touch it, what it costs, and whether it's healthy — in one click, across every catalog, with zero access to your row data.

1 click

from any table to its full upstream & downstream lineage

Column-deep

reconstructs the actual SQL/PySpark that built each column

Metadata-only

reads system tables & BROWSE — never your data

01 The opportunity

Unity Catalog already captures lineage from every operation. The value is locked behind hand-written system-table SQL that only a few engineers can run. BrickTrace unlocks it for the whole organization.

TODAY, WITHOUT BRICKTRACE

- "Where does this number come from?" takes hours of tracing across notebooks, jobs, and SQL.
- A schema change ships and **silently breaks** a downstream dashboard nobody knew existed.
- Nobody can say **how** a column was calculated without reading the pipeline code.
- Sensitive data spreads downstream untracked; access grants pile up unused.
- Pipeline failures and cost spikes are found late, in separate tools.

WITH BRICKTRACE

- Pick any table → the **complete** lineage cone renders instantly, across all catalogs.
- Blast-radius analysis shows exactly **who breaks** before you change anything.
- Click a column → see the real transformation expression that produced it.
- Sensitivity, ownership, and actual-vs-declared access are one panel away.
- Run health, cost, and root cause live **on the graph** — no tool-hopping.

Why BrickTrace wins

- **Goes deeper than the catalog.** Reconstructs the actual expression behind every column — not just "A depends on B."
- **Whole-estate, one click.** Traces across every catalog & schema automatically, with the pipelines in between.
- **Zero-trust by design.** Metadata-only — reads system tables & BROWSE, never a single row of your data.
- **Native & instant to deploy.** A Databricks App — one command, shared across your workspace, no external infra.
- **Scales to thousands.** Shared caching means the warehouse is barely touched, even org-wide.
- **Beyond lineage.** Governance, access, run health, and cost unified in the same view.
- **Speaks to every audience.** One click flips the graph into a plain-language Business view — same lineage for analysts, stewards, and execs, not just engineers.
- **Explains itself.** An AI narrator turns the on-screen graph into a plain-English source → process → output walkthrough.

02 What you get

Seven capability areas, unified in one interactive workspace.

End-to-end lineage, instantly FOUNDATION

Select any table and BrickTrace auto-traces the full lineage cone — every upstream source back to every downstream target, across all catalogs and schemas, with the mediating jobs and pipelines. Switch between Tables, Pipelines, and Full views, cap the depth, and search the graph with Cmd/Ctrl+K.

Business value: onboard to any dataset in seconds and answer "where does this come from and where does it go?" without tribal knowledge or a data-engineering ticket.

Column & transformation lineage DIFFERENTIATOR

Column-level lineage from real Unity Catalog edges — zero false positives. Then the layer no one else has: BrickTrace parses the producing code to reconstruct the **actual** SQL/PySpark expression behind each column — casts, aggregates, CASE logic, window functions, CTE chains — across notebooks, jobs, DLT pipelines, views, and query history. An **AI overview** narrates a table's transformations in plain English; for metadata-driven frameworks whose column logic lives in config tables, a deep agentic pass derives the mappings from the config and parameters the pipeline actually reads.

Business value: prove how any metric was calculated for audits and migrations, and trace a wrong number to the exact line of logic that caused it.

Impact analysis & blast radius GOVERNANCE

Before you change or deprecate a table, see every downstream table it feeds, their owners, the dashboards and jobs that consume it, and whether any of it carries sensitive data.

Business value: ship changes with confidence and prevent the "silent break" that turns into a Monday-morning incident.

Governance, classification & access GOVERNANCE

Ownership and column sensitivity with downstream propagation, custom classification rules that follow the data, and access lineage that joins **declared** grants with **actual** audited usage — surfacing dormant grants to revoke.

Business value: pass access recertification and data-privacy reviews faster, and enforce least privilege with evidence instead of guesswork.

Run health & cost, on the graph OPERATIONS

Every job and pipeline node carries a live cost badge and a health check: a Healthy/Degraded/Failing verdict, success rate, duration trend, and the last runs with per-run cost and direct links. Root-cause tracing walks upstream to the prime-suspect producer when a table goes stale.

Business value: cut mean-time-to-resolution on data incidents and spot the expensive or failing pipelines to fix first — all without leaving the lineage view.

Discovery, quality & AI/ML lineage INSIGHT

Full-text asset search, PII/sensitive-column and orphan-table finders, lightweight per-column data quality rules, and model → training-data lineage that ties UC-registered models back to the tables they learned from.

Business value: retire unused assets, find hidden sensitive data, and know instantly which ML models are exposed when a source table has an issue.

Business view & AI explanations FOR EVERYONE

A one-click **Business view** flips the engineering graph into plain language for non-technical stakeholders: technical types become business terms (**Process**, **Data pipeline**, **Dataset**, **File**), `snake_case` names read as titles, and every dataset carries a plain-English description — while fully-qualified names, columns, IDs, and cost are hidden. A **data-only** mode shows just how datasets connect, using the **precise** Unity Catalog source→target pairs (never a fabricated mesh), and an AI **Explain this lineage** button narrates the whole flow — source → process → output — in a few sentences anyone can read.

Business value: put lineage in front of analysts, stewards, and executives without a data-engineering translator — the same graph serves every audience.

03 Who it's for

One tool, value for every data role.

Data Engineers

BUILD & OPERATE PIPELINES

- Trace breakages to the exact upstream producer, fast.
- See blast radius before shipping a schema change.
- Monitor run health & cost inline.

Data Stewards & Governance

OWN TRUST & COMPLIANCE

- Track sensitive data as it propagates downstream.
- Recertify access with declared-vs-actual evidence.
- Apply classification rules that follow the data.

Analysts & Data Consumers

TRUST & USE THE DATA

- Self-serve "where does this number come from?"
- See exactly how a metric was calculated.
- Judge freshness and reliability at a glance.

Platform & FinOps Leaders

COST, SCALE & RISK

- Find the costliest pipelines in any data flow.
- Deploy org-wide with metadata-only security.
- Scale to thousands of users on one warehouse.

04 Outcomes it drives

Business question	How BrickTrace answers it	Outcome
Where does this data come from and go to?	One-click end-to-end lineage across all catalogs	Faster onboarding & analysis
Who breaks if I change this table?	Impact / blast-radius with owners & consumers	Fewer production incidents
How was this column actually calculated?	Reconstructed transformation expressions	Audit-ready trust
Where is our sensitive data flowing?	Sensitivity classification + downstream propagation	Reduced compliance risk
Who can vs. actually accesses this?	Declared grants joined with audit usage	Least-privilege enforcement
Why is this dataset stale or wrong?	Health-based root-cause tracing to the prime suspect	Lower incident MTTR
What is this pipeline costing us?	Per-node serverless cost & per-run breakdown	Targeted cost optimization
Which ML models used this table?	Model → training-data lineage	Contained model risk
How do I explain this flow to a non-engineer?	One-click Business view + AI plain-English walkthrough	Lineage for every audience

05 Built for Databricks, deployed in minutes

Native, secure, and effortless to roll out

BrickTrace ships as a **Databricks App** (FastAPI + React) — deploy with a single command and it's instantly shared across your workspace. No external infrastructure, no data movement, no Redis or third-party service. It reads only `BROWSE` metadata and Unity Catalog system tables, and writes only to one dedicated app-owned schema — so it clears security review quickly. Request coalescing and shared caching keep warehouse usage minimal even at organization scale.